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Primary productivity of river chaliyar of Calicut district, Kerala, India

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Abstract Productivity is the rate at which inorganic carbon is converted into organic carbon in an ecosystem. The present study was carried out to study the status of fresh water Chaliyar River, Calicut, Kerala. Primary productivity was assessed by estimating the changes in dissolved oxygen concentration by "Light and Dark bottle method" on monthly basis for a period of six months (Aug-2012-Jan-2013).GPP values in the river ranged from 0.90 – 3.95 mg C/l/day and NPP values in the river varied from 0.13 to 0.84 mg C/l/day. Thus, it is apparent that values recorded from different stations of the river in general and at station II and III in particular are much higher than those prescribed for a healthy ecosystem. Therefore, the present investigation has been carried out with the objectives to assess the primary productivity of Chaliyarriver to assess the environmental condition and health of the water body.

Key Words Primary productivity, Chaliyarriver, Net production, gross production.

Introduction

The primary productivity of a fresh water body is the manifestation of its biological production and sources of energy input (Verma and Mohanti, 1994) in aquatic ecosystem. Primary productivity is directly related to the temperature, available nutrients in water, soil and in relation to other physicochemical factors (Moss *et al*., 1980). The importance of study of primary productivity of an aquatic ecosystem lies in the fact that firstly it gives valuable information on the productivity values and also serve as an index of pollution status of water body (Harikrishnan et al., 2000), the diversity in the distribution, abundance and variations in the biotic factors provide information of energy turn over in the aquatic ecosystem **(**Forsbery**,** 1992**).** Chaliyar River is important for recreation, domestic water supply, transportation and agricultural purposes particularly cassava processing for the production of cassava flour and other products. Increasing problem of deterioration of river water quality, it is necessary to monitoring of water quality to evaluate the production capacity. Thus, the acute societal and global need to monitor the quality and characteristics of water bodies, upon which many rural and urban communities depend for survival, cannot be neglected. Therefore, the present investigation has been carried out with the objectives to assess the monthly variation in primary productivity of Chaliyarriver at the selected stations with a view to find out some clues for river water management.

Materials and methods

Study Area: Chaliyar is the fourth longest river in Kerala at 169 km in length. The Chaliyar is also known as Chulika River or Beypore River as it nears the sea. The main river starts from the Elambalari hills at an altitude of 2,067 m above MSL. It is formed by the confluence of numerous streams and rivers. The Chaliyar, flowing for a total length of about 169 km finally joins the Arabian Sea

at Beypore. The river drains a total area of about 2,933 Sq km. of which 388 Sq km. liesin Tamil Nadu. Nilambur, Edavanna, Areekode, Kizhuparamba, Cheruvadi, Mavoor, Peruvayal, Feroke and Beypore are some of the towns/villages situated along the banks of Chaliyar. Chaliyar River originates in the Western Ghats range at Elambalari Hills located near Cherambadi town in the Nilgiri district of Tamil Nadu and flows through Malappuram District for most of its length and then for around 17 Kms it forms the boundary between Malappuram District and Kozhikode District before entering the city of Kozhikode for its final 10 Kms journey and finally empties into the Arabian Sea.

Sampling Sites: Three sampling stations were established after a reconnaissance visit to the river. The criteria for the choice of the three sampling stations I (Fig: 2**)**, II (Fig: 3**)** and III (Fig: 4**)** along the river course based were chosen on disturbance level and as non-dredged station dredged station and partially dredged station.

Coordinates: 11°14'49"N 75°57'22"E

Coordinates: 11.25274°N 75.96097°E

Coordinates: 11°15'18"N 75°56'26"E

Primary productivity: Primary productivity was assessed by estimating the changes in dissolved oxygen concentration by "Light and Dark bottle method" (Gaarder and Gran, 1927). All the reagents used for the analyses were of analytical grade and obtained from Merck Germany.

Procedure

Collect water samples always in the series of 'L', 'D' and 'I' labelled 125 or 250 ml BOD glass bottles without entangling air bubbles and close the lids gently. The filtered water samples thus collected are set for incubation atleast 30 minutes (30 minutes to one hour) after sampling to bring the microorganisms physiologically stabilized inside the 'I', 'D' and 'L' bottle. Fix the 'I' bottle with Winkler 'A' and 'B' reagents and keep the 'L' bottle exposed to uniform light and 'D' bottle in darkness (enclosed in dark bags or wrap with thick black cloth or any similar material) in the Laboratory or Research Vessel under simulated *in situ* environment at room (normal) temperature during incubation time. Fix 'L' and 'D' bottles after incubation time with Winkler 'A' and 'B' and determine dissolved oxygen values upto three decimal points for 'I', 'D' and 'L' bottle samples.

Calculation

GPP = (L-D) value in ml O_2/l per 12 light hours per day NPP = (L-I) or 0.8 (L-D) value in ml O_2/l per 12 light hours per day (whichever is applicable). For uniformity, 0.8 (L-D) per 12 hrs may be considered for NPP. ***Conversion factor***: 1 ml O_2 released by photosynthesis is equivalent to 0.536 mg C. Average photosynthetic Quotient (PQ) =1.25. Hence, 1 ml O_2 = 0.536 = 0.429 mg. C. 1.25 GPP or NPP value of oxygen (ml O_2/l/D) multiplied by 0.429 would give the Primary (Photosynthetic) Productivity value in mg. C/l/d.

Results and discussion

The results of the mean values of the primary productivity (Net productivity, Gross productivity and respiratory action) of the selected three sampling sites- I,II and III of study area Chaliyar River were represented in Table:1-3.

Primary productivity is the capacity of the ecological aquatic ecosystem, community or any part is defined as the rate to convert radiant energy to chemical energy via primary transducers is stored, by photosynthetic and

chemosynthetic activities of producer organisms, in the form of organic substances which can be used as food materials and its subsequent transfer to higher levels of the ecosystems via primary consumers. The assessment of primary productivity of freshwater water bodies provides information pertaining to photosynthetic production of organic matter and other energy related functional aspects of an aquifer. It has been a reliable index of pollution assessment (Wetzel and Linkers 1979; Christi *etal* ., 2011) as it is the back bone of all food chains and food – webs of an ecosystem and generates 70% of the world's atmospheric oxygen (Reynolds, 1984). Primary productivity depends on a number of factors such as availability of nutrients, quality, quantity and duration of light, transparency or euphotic zone of the water body and the presence or absence of inhibitors or toxicants in the ambient water.

Changes and differences in quality and quantity of basic productivity within a river may be expected to influence, although not always directly, the quality and quantity of organic accumulation in successive trophic levels. Thus, the present study revealed that river water of the study area site – I was not polluted in respect to physicochemical assessment when compared to site- II and site –III. But all the sampling locations of the river are subjected to human interference regularly. GPP (Gross primary productivity) values in the river ranged from 0.90 – 3.95 mg/ C/l/day. The maximum value was noticed at station I and minimum at station III in the months of April and September respectively. NPP values in the river varied from 0.13 to 0.84 mg C/l/day. High total solid concentration, low transparency and high turbidity as a consequence of ash discharged from nearby small and large scale industries highly reduced the euphotic zone of the river at stations. Besides this, heavy metals present in the industrial and domestic wastes also seem to exert a negative pressure on the growth, development and population of primary producers. All these factors probably acted in synergy to reduce the growth and development of primary producers at this station. Sewage a rich source of plant nutrition (N and P) stimulating the plant growth is also discharged through the drain but the despite that primary producer's population was very meek due to factors mentioned above.

According to (Ketchen et al., 1958) in a healthy aquatic ecosystem, respiration should be 5 – 10 % of the gross production. Thus, it is apparent that values recorded from different stations of the river in general and at station II and III in particular are much higher than those prescribed for a healthy ecosystem (Ketchen et al., 1958). This indicates an and healthy disturbed ecosystem with low NPP draining more dissolved oxygen than producing it photo – synthetically. GPP should approach unity in a healthy eco system. As per the data obtained in the present study, it may be concluded that water quality of the river slightly improved downstream. The factors that propped up the growth of primary producers and the improved production include water temperature in the bio-kinetic range, ample nutrients (N and P) and low concentration of toxic materials due to natural dilution – as the river recouped in due course of time. All these factors enhanced the productivity at this station as compared to other stations. Taking this value into account, it was concluded that the whole river stretch investigated was highly polluted. Weather conditions also affected the productivity.

Low primary productivity may be attributed to high current velocity, turbulence and turbidity in the river due to run off water

from the catchment areas reducing water transparency and there by euphotic zone of the river leading to reducing light intensity. All these adversely affected the primary producers population and primary productivity as well .high current velocity constantly washed away the producer organisms. Thus, the already meek plankton community got disturbed and productivity decline d significantly during that period. The finding was in conformity with Ahmed *et al*. (2005); emphasized the effect of phytoplankton population on productivity and reported that phytoplankton abundance and productivity correlate significantly. The maximum productivity in summer as reported in the present investigation was in variance with (Kar et al., 1987) who reported maximum production in river in monsoon but was in concordance (Patra, 1985).

The present study corroborates the findings of these authors. It was thus evident that despite optimum temperature, surplus nutrients and bright sunshine, the productivity of the river was quite low due to anthropogenic stress leading to potential toxic condition resulting from the discharge of huge industrial and municipal wastes into the river and that too directly sans any pre-treatment. River was grossly polluted and required special heed from government and nongovernmental organizations so that it can recuperate and survive as a healthy natural ecosystem. It should not be allowed to be converted into a waste depository to carry the anthropogenic wastes beyond its assimilating capacity.

The study attributed that river water is not fit for direct utilization due to the above human interferences, which require continuous monitoring and treatment process if the water is to be used for drinking purposes. Some steps and awareness programs are needed to educate local villagers to safeguard the precious river and its surrounding from further deterioration. It is also clear that variations in pre monsoon are not remaining constant for minimum and maximum concentration during post monsoon. Post monsoon characteristics of river water when compared with pre monsoon shows increased concentration of electrical conductivity, total dissolved solids, total hardness, chloride, phosphate and biological oxygen demand. This fluctuation may be attributed by the activities like discharge of sewage in the river and decreased and periodical flow of water in the river after monsoon season.

Conclusions

Thus, the present study revealed that river water of the selected study area sites was highly polluted except Site-I which is least polluted. But all the sampling locations of the river are subjected to human interference regularly. The study attributed that river water is not fit for direct utilization due to the above human interferences, which require continuous monitoring and treatment process if the water is to be used for drinking purposes. Some steps and awareness programs are needed to educate local villagers to safeguard the precious river and its surroundings from further deterioration. The study concluded that there were significant monthly variations in all parameters under study. From the above discussion and the result analysis in different aspects it may be concluded that Chaliyar river water quality is **not good** mostly during post monsoon season which means that the anthropogenic influences such as urbanisation and agricultural activities are the main cause for deterioration of water quality. The overall results of the present study indicate that in Chaliyar River the major cause of pollution is

the population density and obviously the discharge of domestic sewage from point and non-point sources and the industrial effluents also contribute to its population load.

Table 1: Data on the monthly variations of primary productivity of river chaliyar at site-I

Months	Primary productivity (mg C/l/day)		
	Net production	Gross production	NP/GP (RA)
August	1.66	2.86	0.58
September	1.70	2.95	0.58
October	1.90	3.15	0.60
November	2.53	3.95	0.64
December	1.80	3.20	0.56
January	2.08	3.40	0.61

*All the values are means of triplicate observations (N=3)

Table 2: Data on the monthly variations of primary productivity of river Chaliyar at site-II

Months	Primary productivity (mg C/l/day)		
	Net production	Gross production	NP/GP (RA)
August	0.75	1.90	0.39
September	0.85	2.05	0.41
October	0.76	1.96	0.39
November	1.47	2.70	0.54
December	1.03	2.20	0.47
January	1.13	2.25	0.50

*All the values are means of triplicate observations (N=3)

Table 3: Data on the monthly variations of primary productivity of river Chaliyar at site-III

Months	Primary productivity (mg C/l/day)		
	Net production	Gross production	NP/GP (RA)
August	0.80	1.90	0.42
September	0.78	1.90	0.42
October	0.69	1.83	0.38
November	0.20	1.50	0.13
December	0.76	0.90	0.84
January	0.63	1.75	0.36

*All the values are means of triplicate observations (N=3)

Fig. 1: River Chaliyar

Fig. 2: Site-I Mananthalakadavu

Fig. 3: Site-II : Elamaram

Fig. 4: Site-III: Kalpally

References

Ahmed, T. (2005). Primary production and fish yield estimation in the meghana river system, Bangladesh; Asian Fisheries Science, 18: 95-105.

Forsberg, C. (1982). Limnological research can improve and reduce the cost of monitoring and control of water quality. Hydrobiol. 86: 143-146.

Gaarder and Gran, (1927). Investigation on the production of plankton in the Osaio - Fiiord- Rapp- Cons. intt. Explor. Mer 42, 1-48.

Harikrishnan, K., and Abdul Azis. (2000). Primary productivity studies in a freshwater temple tank ' Kerala. Ind. J. Environ & Ecoplan., 3(1): 127-130.

Kar, G.K., Mishra, P.C., Dash, M.C. and Das, R.C. (1987). “Polution studies in river Ib III: Plankton Population and Primary Productivity; Ind. J. Environ. Hlth., 29:32-39.

Ketchen, B.H., Ryther, J.H., Yentsch, C.S. and Corwin, N. (1958). Productivity in relation to nutrients. Rapp. P.V. Renu. Cons. Per. Int. Explor. Mer., 144: 132-140.

Moss, B., R. G. Wetzel and G. H. Luff 1980. Annual productivity and phytoplankton changes between 1969 and 1974 in Gull Lake, Michigan. Freshwater Biol., 10: 113-121.

Patra, R.W.R. and Azadim.A. 1985. Limnology of Halda River. The Journal of NOAMI **2**(2): 31-38.

Reynolds, C.S. (1984). The ecology of fresh water phytoplankton. Cambridge University. Prss, Cambridge and Newyork. 384.

Verma, J.P. and R.C. Mohanty (1994). Evaluation of water quality of two fresh water ponds. Pollut. Res.13:69-74.

Wetzel, R. G., and Likners, G.E. (1991). Limnological analysis - 2nd eel. Springer Newyork.